

'Service' corresponds to the list of services that we specified in the `/etc/services` file (see the previous column for more information). 'Socket_type' is used to specify the relevant socket for the 'service' entry (raw, rdm, etc) and 'proto' will indicate the valid protocols to be used for this (tcp, udp, etc).

'Flag' can be set to instruct the daemon whether or not to free a socket after its use by a connection. You have to look at your requirements list and the protocols used, in order to decide the value for this field.

The 'User' field assigns a particular user in the system (from `/etc/passwd` file) as the owner of the network daemon. It is a security related entry as you are restricting the access rights and privileges using this.

Typically, the root user will be shown as the owner of the network daemon. You can use 'server_path' to provide the pathname to the actual server program and 'server_args' (optional entry) to pass an argument to the daemon program.

`/etc/ftptusers` is another important file. It is used to deny access to certain users (of the system) from using the ftp service. The daemon program (`ftpd`) looks for the entries in this file, every time it receives a request for a connection. Here is a sample entry:

```
# /etc/ftptusers - users not allowed to login via ftp
root
uuucp
bin
mail
```

You can find an entry corresponding to the 'root' user in the above code. This is done due to security reasons.

By using the `/etc/securetty` file, you can specify which *tty* devices are open for the root to log in. Here is a typical entry:

```
# /etc/securetty - tty's on which root is allowed to login
tty1
tty2
tty3
tty4
```

You need to be aware that this is also security related and you should be careful while meddling with this file.

The `/etc/hosts.deny` file is yet another configuration file. This is used by the `/usr/sbin/tcpd` program to find the hosts that are disallowed from using a particular service. Here is a valid entry for the same:

```
# /etc/hosts.deny
#
# Disallow all hosts with suspect hostnames
ALL: PARANOID
#
```

```
# Disallow all hosts.
ALL: ALL
```

Avahi daemon

We have looked at some of the Avahi-related points earlier. As we discussed, it is a daemon that helps in service discovery on a local network. This essentially lets you to connect your laptop to a network and find other systems/services in the network. It is based on Lennart Poettering's mDNS implementation (called FlexMDNS) for Linux.

Here is the code portion that handles the enabling and disabling processes:

```
enable_avahi () {
# no unicast .local conflict, so remove the tag and start avahi again
if [ -e $(DISABLE_TAG) ]; then
rm -f $(DISABLE_TAG)
start avahi-daemon || :
fi
}

disable_avahi () {
[ -e $(DISABLE_TAG) ] && return

stop avahi-daemon || :
if [ -x /usr/bin/logger ]; then
logger -p daemon.warning -t avahi <<EOF
Avahi detected that your currently configured local DNS server serves
a domain .local. This is inherently incompatible with Avahi and thus
Avahi disabled itself. If you want to use Avahi in this network, please
contact your administrator and convince him to use a different DNS domain,
since .local should be used exclusively for Zeroconf technology.
For more information, see http://avahi.org/wiki/AvahiAndUnicastDotLocal
EOF
fi
ensure_rundir
touch $(DISABLE_TAG)
}
```

The daemon is basically a free Zeroconf implementation (based on the Apple Zeroconf specification) and implements mDNS, DNS-SD and RFC 3927/IPv4LL. You could find this in almost all Linux flavours, and it also provides a set of language bindings. Avahi has already been incorporated into GNOME's Virtual File System (VFS) and KDE's input/output architecture.

We have covered most of the basic networking aspects. Till the next issue, Happy Kernel Hacking! 

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